

Chapter 31: Mithun (*Bos frontalis*): A Review on Feeding Ecology, Natural Fodder Resource Management, Habitat and Conservation in the Eastern Himalaya

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Abstract

Mithun (Bos frontalis), commonly known as gayal or the “cattle of the mountains,” is a unique semi-domesticated bovine closely associated with the forest ecosystems of the Eastern Himalaya. This review synthesizes available literature on the feeding ecology, natural fodder resources management, habitat relationships, and conservation of Mithun, with particular emphasis on north-eastern part of India and adjoining regions. Mithun exhibits predominantly browsing behaviour and utilizes a highly diverse range of trees, shrubs, herbs, grasses, climbers, creepers, bamboo and other seasonal plant resources. Studies have reported fodder species with leaves and young shoots forming major dietary components. Feeding behaviour is strongly influenced by vegetation availability, seasonality, climatic conditions, water sources and habitat structure. Natural forests therefore play a critical role in sustaining Mithun under traditional extensive production systems. However, deforestation, shifting cultivation, agricultural expansion, infrastructure development, habitat fragmentation and climate change are increasingly threatening fodder availability and suitable habitats. Conservation of preferred fodder species, fodder plantations, agroforestry, nutritional supplementation and semi-intensive management can improve forage security and productivity. Integrating indigenous ecological knowledge with scientific approaches is essential for sustainable Mithun production, biodiversity conservation and long-term preservation of this culturally and ecologically important livestock resource of the Eastern Himalaya.

Keywords: *Bos frontalis, Mithun, feeding ecology, fodder; indigenous knowledge, Eastern Himalaya, conservation*

1. Introduction

Mithun (*Bos frontalis*), commonly known as the Gayal or the Drung ox, also frequently referred to as the "Cattle of the Mountains" due to its habitat in the hilly forests of the Eastern Himalaya, is one of the most unique semi-domesticated bovine species found in the Eastern Himalayan region. It is predominantly distributed across the states of Arunachal Pradesh, Nagaland, Manipur, and Mizoram in India, with smaller populations occurring in Bhutan, Myanmar, Bangladesh, and China. Unlike common domestic cattle, Mithun has evolved under free-ranging conditions within tropical and subtropical forest ecosystems, where it depends almost entirely on naturally available vegetation for its nutritional requirements. Owing to its exceptional adaptability to rugged mountainous terrain, superior meat quality,

cultural significance, and ecological importance, Mithun occupies a distinctive position among indigenous livestock species of South and Southeast Asia (Lambert., 1804; Gupta et al., 1996; Mondal et al., 2014; Kumar et al., 2025).

For centuries, indigenous tribal communities have reared Mithun under extensive forest-based production systems with minimal external inputs. They are generally allowed to roam freely through forest landscapes, browsing on a diverse range of grasses, shrubs, herbs, climbers, bamboo, and tree foliage while receiving little or no supplementary feeding except for periodic mineral salt supplementation (Giasuddin & Islam, 2003; Shisode et al., 2009). This traditional husbandry system reflects a close interaction between the species and its natural habitat, making the availability and diversity of forest vegetation a critical determinant of animal health, productivity, and survival. Consequently, the ecological sustainability of Mithun husbandry is inseparable from the conservation of forest ecosystems that provide food, water, shelter, and breeding habitats.

Scientific interest in Mithun has increased considerably owing to its growing economic potential, nutritional importance, and conservation value. Research has grown beyond descriptive studies on morphology and distribution to encompass diverse aspects including taxonomy, genetics, reproductive biology, nutritional physiology, behavioural ecology, meat science, conservation biology, and sustainable livestock production (Ahrestani, 2018; Kumar et al., 2025; Dorji et al., 2021). Significant progress has also been made in understanding the nutritional requirements of Mithun, evaluating indigenous fodder resources, developing semi-intensive feeding systems, and improving productivity through scientific management practices (Prakash et al., 2008; Das et al., 2011; Sinha et al., 2011). These studies collectively demonstrate that although Mithun possesses remarkable adaptability to forest environments, its productivity can be substantially improved through enhanced nutritional management while preserving the ecological characteristics that define the species.

Among the various research themes, the feeding ecology of Mithun has emerged as one of the most important areas of investigation. Feeding behaviour determines not only nutrient intake and animal productivity but also influences habitat utilization, movement patterns, forest regeneration, biodiversity conservation, and the sustainability of traditional livestock production systems. Unlike grazing cattle that rely primarily on pasture grasses, Mithun exhibits a predominantly browsing habit, utilizing an exceptionally diverse array of forest vegetation including tree leaves, shrubs, herbs, climbers, bamboo shoots, grasses, and seasonal fruits (Taba et al., 2015; Geng et al., 2017; Ahrestani, 2018). The remarkable dietary flexibility of the species enables it to survive under heterogeneous forest conditions and fluctuating seasonal availability of forage resources. The understanding of feeding behaviour is therefore fundamental for designing appropriate nutritional interventions, conserving preferred fodder species, and ensuring the long-term sustainability of Mithun husbandry.

Recent investigations have considerably enriched scientific knowledge regarding the diversity of fodder species consumed by Mithun. Surveys conducted in India and China have documented between 40 and 142 naturally occurring forage species comprising trees, shrubs, herbs, grasses, climbers, creepers, and bamboo, demonstrating the extensive dietary breadth of the species (Geng et al., 2017; Jamir & Khare, 2018; Geng et al., 2020; Taba et al., 2015). These studies have highlighted the importance of integrating indigenous ecological knowledge with scientific investigations, as local tribal communities possess

detailed traditional understanding of fodder availability, seasonal browsing patterns, and preferred forage species accumulated over generations of experience.

Despite these advances, important knowledge gaps remain in the understanding of Mithun feeding ecology. Most available studies are confined to specific geographical regions, while comprehensive comparative assessments of fodder diversity across different ecological zones remain limited. Increasing anthropogenic pressures such as deforestation, shifting cultivation, expansion of agricultural land, infrastructure development, and climate change have further reduced natural grazing habitats, emphasizing the urgent need for scientific documentation and conservation of Mithun fodder resources (Faruque et al., 2015; Chavan et al., 2018; Kumar et al., 2025).

The present review of literature aims to synthesize the existing body of knowledge on Mithun with particular emphasis on its feeding ecology and fodder diversity. Previous studies relating to the origin, distribution, ecology, husbandry, feeding behaviour, natural fodder resources, conservation issues, and sustainable management strategies have been critically examined to identify the current state of knowledge and existing research gaps. This review provides the conceptual and scientific foundation for the present investigation and establishes the rationale for undertaking further research on the fodder species and feeding habits of Mithun under natural forest conditions. It also serves to position the present study within the broader context of sustainable Mithun conservation and scientific livestock management.

2. Literature sources and selection

This review synthesized scientific literature on the Mithun (*Bos frontalis*) retrieved from electronic databases including Google Scholar, PubMed, Scopus, and CAB Direct, alongside official technical reports from the ICAR-National Research Centre on Mithun. Search queries combined primary terms (*Bos frontalis*, *Mithun*, *Gayal*) with thematic descriptors (*feeding behaviour*, *foraging patterns*, *natural fodder/food plants*, *fodder diversity*, *fodder resources* etc.). Peer-reviewed articles and validated institutional data detailing Mithun feeding, ecology, feeding behaviour, fodder resource management, and conservation status were selected for critical synthesis, while duplicate records and non-scientific sources were excluded.

3. Origin and distribution

The taxonomic origin of Mithun has long been debated among zoologists and livestock researchers. The binomial '*Bos frontalis*' was first applied in 1804 by Lambert. Studies suggested that Mithun descended directly from the wild gaur (*Bos gaurus*), while others proposed that it originated through hybridization between gaur and indigenous cattle. Morphological similarities in body structure, horn configuration, coat colour, and skeletal characteristics strongly support its close evolutionary relationship with the gaur. Modern molecular investigations have further indicated that Mithun shares substantial genetic affinity with gaur, although centuries of semi-domestication under human management have produced distinct behavioural and phenotypic characteristics that justify its recognition as a separate domesticated bovine lineage (Gupta et al., 1996; Mondal et al., 2014). Despite continuing discussions regarding its exact evolutionary origin, there is broad consensus that Mithun represents a unique genetic resource possessing exceptional adaptive traits that are highly valuable for livestock biodiversity conservation.

The geographical distribution of Mithun is confined primarily to the humid mountainous regions of South and Southeast Asia. India supports the largest global population, with the species occurring

predominantly in Arunachal Pradesh, Nagaland, Manipur, and Mizoram. Smaller populations are also found in Bhutan, Myanmar, Bangladesh, and China (Mondal et al., 2014; Faruque et al., 2015; Wangchuk et al., 2021). Within India, Arunachal Pradesh possesses the highest Mithun population and is widely recognized as the principal centre of Mithun husbandry. The species is generally distributed between elevations of approximately 200 and 3,000 metres above sea level, although the most suitable habitats occur in tropical and subtropical evergreen forests characterized by moderate temperatures, abundant rainfall, high relative humidity, and perennial freshwater sources (Taba et al., 2015).

4. Ecology and Traditional Husbandry

The ecological adaptations of Mithun reflect its long evolutionary association with forest ecosystems. Unlike domestic cattle that thrive in open grasslands and cultivated pastures, Mithun is essentially a forest-dependent browser capable of exploiting a wide diversity of natural vegetation. Dense tropical forests provide not only food resources but also shelter from solar radiation, protection against predators, and favourable microclimatic conditions necessary for thermoregulation. Studies have consistently reported that Mithun prefers humid environments with annual rainfall exceeding 2,000 mm, relative humidity often ranging between 80 and 90 percent, and dense vegetation interspersed with streams, rivers, springs, and salt licks (Giasuddin & Islam, 2003; Taba et al., 2015). Because of its comparatively sparse body hair and dark coat colour, Mithun experiences rapid heat absorption under direct sunlight and consequently restricts most feeding activities to early morning, late evening, nighttime, or rainy periods. During hot daytime hours, animals usually remain under dense forest canopy or near water bodies where environmental temperatures remain comparatively lower.

Another remarkable ecological characteristic of Mithun is its exceptional mobility within rugged mountainous terrain. The species routinely traverses steep slopes, dense forests, river valleys, and bamboo thickets while searching for preferred browse species. Such mobility allows Mithun to utilize spatially dispersed food resources and adapt to seasonal fluctuations in forage availability. Seasonal movement patterns are closely associated with changes in vegetation phenology, water availability, and climatic conditions. During periods of fodder scarcity, animals often expand their foraging range to locate nutritious young foliage, whereas during seasons of abundant vegetation they remain within relatively smaller home ranges. These behavioural adaptations enable Mithun to maintain nutritional balance despite living under extensive free-ranging conditions where forage resources are naturally heterogeneous.

The relationship between Mithun and indigenous tribal communities extends far beyond conventional livestock production. Throughout Northeast India, Mithun occupies an extraordinary position within the socio-cultural, economic, and religious life of numerous tribes. It serves as a symbol of wealth, prestige, social status, and cultural identity. Ownership of large Mithun herds traditionally reflects economic prosperity and social influence within tribal society. They are frequently exchanged during marriage negotiations, presented as gifts during festivals, sacrificed in important religious ceremonies. (Mondal et al., 2014; Jamir & Khare, 2018). Unlike commercial livestock raised primarily for market production, Mithun therefore represents both a biological resource and a living cultural heritage whose value extends beyond its economic contribution.

Traditional Mithun husbandry has remained largely unchanged for generations and is characterized by extensive free-range management systems. Animals are generally released into surrounding forests where

they roam freely throughout the year, subsisting almost entirely on naturally occurring vegetation with minimal human intervention. Owners usually identify individual animals through ear markings, branding, or distinctive horn characteristics, periodically locating them within forest habitats for health monitoring or ceremonial purposes. Supplementary feeding is uncommon except during periods of severe fodder shortage, while mineral supplementation is typically provided through natural or artificial salt licks established within forest areas. Such low-input production systems substantially reduce feeding costs but simultaneously increase dependence upon healthy forest ecosystems and naturally available fodder resources (Shisode et al., 2009; Mondal et al., 2014; Taba et al., 2015).

5. Feeding Behaviour and Foraging Ecology of Mithun

Mithun's feeding behaviour is very important for animal nutrition, productivity, habitat utilization and the survival of traditional forest-based production systems. Mithun (*Bos frontalis*) is highly adaptable to climate and habitat conditions in humid subtropical and tropical forests in the eastern Himalayas and adapts its diets to seasonal forage availability (Giasuddin & Islam, 2003; Taba et al., 2015). Studies from India, Bangladesh and China have shown that Mithun can use a wide variety of natural plant resources. One of the most distinctive characteristics is its browsing behaviour. It constantly traverses forest habitats to find suitable fodder species and selects leaves, twigs, young shoots, herbs, bamboo foliage and climbers rather than feeding indiscriminately (Giasuddin & Islam, 2003). Fresh, tender and green plant parts are generally preferred, whereas dried, wilted, diseased or decaying vegetation is avoided (Taba et al., 2015). This selective feeding is a crucial ecological adaptation that enables Mithun to obtain nutritionally valuable food from diverse forest vegetation. Mithun also has a clear preference for certain parts of plants. Jamir and Khare (2018) found that leaves made up about 70% of the diet, followed by whole plants (20%), twigs (7-8%) and shoots (around 4%). Taba et al. (2015) identified leaves as the most important food source among 42 fodder species. The preference for young foliage is nutritionally significant because tender leaves may have much more protein, minerals, and digestible nutrients than mature woody tissues. The wide variety of plants consumed also reflects Mithun's feeding strategy. Geng et al. (2017) found more than 100 fodder species in the Dulongjiang region of China, while Taba et al. (2015) recorded 42 fodder species in Arunachal Pradesh and Jamir and Khare (2018) identified 39 food species in protected areas in Nagaland. These include trees, shrubs, herbs, grasses, climbers, creepers and bamboo. This diversity of diet helps Mithun to access a variety of different locations and seasonally different resources to forage and reduces reliance on a single plant species. Feeding is a significant part of the daily activity budget of free-range Mithun. Jamir and Khare (2018) found that feeding accounted for approximately 61.66% of daytime activity in the field, but there are wide differences across habitat types. Feeding accounts for 77% of daytime activity in Intanki National Park but only about one-third in Dzuleke, where animals moved longer distances between feeding and resting areas. So the vegetation density, forage distribution and habitat quality affect feeding time and movement. Climate also impacts feeding activity. Mithun usually avoids intense solar radiation and is more active during cool periods such as early morning, late evening, and rainy days. In hot afternoons, animals are usually under forest canopy or close to water sources (Giasuddin & Islam, 2003; Taba et al., 2015). Thus thick evergreen and subtropical forests offer both forages and favourable microclimatic conditions for foraging. Water sources such as rivers, streams, springs, and ponds also affect habitat choice and foraging opportunities. Mineral-seeking behaviour is also an important aspect of Mithun feeding ecology. Animals often visit natural salt licks and mineral-rich soils, and indigenous people offer salt at designated locations to supplement mineral requirements (Giasuddin & Islam, 2003). This is a perfect example of the integration

of natural foraging behaviour with traditional husbandry. Seasonal changes in vegetation availability can also influence feeding behaviour. When it rains, the vegetation has abundant forage in the form of fresh leaves, herbs, climbers and bamboo shoots, and in drier seasons Mithun can increase its foraging range and use mature leaves, grasses and woody browse. While quantitative studies on seasonal dietary composition remain limited. In general, the feeding ecology of Mithun is characterized by the interaction of dietary flexibility, habitat characteristics, climate, seasonal forage availability and indigenous management practice. Integrating traditional ecological knowledge with scientific nutritional and ecological research is crucial for sustainable feeding and conservation approaches.

6. Habitat Conservation and Fodder Resource Management

Mithun naturally inhabits dense evergreen and semi-evergreen forests with high rainfall, moderate temperatures, perennial water sources and diverse vegetation. Dense canopy provides shade and favourable microclimatic conditions, while streams, springs, rivers and natural salt licks meet water and mineral requirements (Giasuddin & Islam, 2003; Taba et al., 2015). Thus, suitable habitat depends not only on vegetation availability but also on water resources, forest structure, topography and climatic conditions.

Vegetation diversity is a major determinant of habitat quality. Forests provide trees, shrubs, herbs, grasses, climbers and bamboo that collectively support the nutritional requirements of free-ranging Mithun. Taba et al. (2015) recorded 42 fodder species in Arunachal Pradesh, while Geng et al. (2017) documented approximately 142 species in Dulongjiang, China. Such diversity helps maintain forage availability despite seasonal variation. Consequently, forest degradation and loss of vegetation diversity can directly reduce forage availability and nutritional security.

Mithun habitats are increasingly threatened by agricultural expansion, shifting cultivation, logging, infrastructure development, mining, urbanization and settlement expansion. Habitat fragmentation restricts animal movement and access to preferred fodder and water resources. Smaller forest patches may also contain fewer nutritionally important browse species and increase human–animal interactions. Faruque et al. (2015) reported that declining forest cover in Bangladesh reduced grazing areas and contributed to shrinking Mithun populations.

Climate change may further affect habitat suitability through altered rainfall, increasing temperatures, prolonged dry periods and changes in vegetation composition. Such changes can influence forage quantity, quality and seasonal availability. Increased temperatures may also reduce daytime feeding activity and increase physiological stress. Therefore, future conservation strategies should incorporate climate adaptation along with habitat protection.

Conservation of natural fodder resources is consequently essential for sustainable Mithun management. Geng et al. (2020) demonstrated that several indigenous browse species possess high nutritional value and digestibility. Protection and regeneration of preferred fodder species, regulation of forest extraction and establishment of protected browse areas can therefore improve forage security. Agroforestry systems incorporating fodder species such as *Ficus*, *Bauhinia*, *Artocarpus*, *Mallotus* and *Pueraria* can provide supplementary forage while reducing pressure on natural forests.

Indigenous communities have an important role in conservation because of their traditional knowledge of forest ecology, seasonal feeding patterns, preferred fodder plants and sustainable livestock management.

Integrating this knowledge with scientific conservation approaches can support locally appropriate management strategies (Geng et al., 2017). Institutional initiatives, including research and conservation programmes of the ICAR–National Research Centre on Mithun and state agencies, further emphasize fodder development, habitat improvement and sustainable management. Overall, conservation of Mithun and conservation of its natural forest habitat should be treated as interconnected objectives.

7. Conclusions

In the long term, the sustainability of Mithun (*Bos frontalis*) will depend on keeping the ecological conditions that support its traditional forest-based production system and adapting management practices with environmental and socio-economic considerations. We must therefore focus conservation efforts on safeguarding and restoring Mithun habitats, particularly in areas where fodder species with valuable and nutritionally important natural fodder are important. Forage plantations and agroforestry based on local browse species are a good way to supplement foraging in the absence of a suitable forage crop, reduce the pressure on natural forests and help boost nutritional security in a time of seasonal scarcity. Nutritional support and semi-intensive production systems that are well-designed to ensure productivity improvement can be achieved in large part with appropriate diets and the species' typical ecological adaptations that contribute to the species' existence.

Future management should be a multi-disciplinary approach integrating scientific livestock management with indigenous ecological knowledge. Traditional knowledge on fodder species, seasonal availability, animal movement, salt-lick sites and forest resource usage should be documented and integrated into conservation and management programs. More needs to be done to build community participation in habitat conservation, fodder development in a climate-resilient manner and sustainable use of forest resources. Future research should focus on long-term assessment of seasonal fodder availability, nutritional quality, feeding preferences and habitat utilization and the impact of climate change on Mithun-forest interactions. In this way, such efforts can provide an evidence-based basis for local conservation strategies. In the end, Mithun conservation will require the management of fodder resources, forest habitat and traditional knowledge systems to be balanced and in both a sense that Mithun will be able to remain a sustainable livestock resource and maintain its ecological and cultural significance to the Eastern Himalaya to maintain our wildlife.

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References

- [1] Ahrestani, F. S. (2018). *Bos frontalis* and *Bos gaurus* (Artiodactyla: Bovidae). *Mammalian Species*, 50(959), 34–50. <https://doi.org/10.1093/mspecies/sey004>
- [2] Chavan, S., Yuvaraja, M., & Sarma, H. N. (2018). *Bos frontalis* (Mithun): A flagship semi-domesticated mammal and its potential for ‘Mithun husbandry’ development in Arunachal Pradesh, India. *Concepts of Dairy & Veterinary Sciences*, 1(5), 134–138. <https://doi.org/10.32474/CDVS.2018.01.000122>

- [3] Das, K. C., Haque, N., Baruah, K. K., Rajkhowa, C., & Mondal, M. (2011). Comparative nutrient utilization, growth, and rumen enzyme profile of Mithun (*Bos frontalis*) and Tho-tho cattle (*Bos indicus*) fed on tree-leaves-based ration. *Tropical Animal Health and Production*, 43(1), 209–214. <https://doi.org/10.1007/s11250-010-9676-1>
- [4] Dorji, T., Wangdi, J., Yi, S., Chettri, N., & Wangchuk, K. (2021). Mithun (*Bos frontalis*): The neglected cattle species and their significance to ethnic communities in the Eastern Himalaya—A review. *Animal Bioscience*, 34(11), 1727–1738. <https://doi.org/10.5713/ab.21.0020>
- [5] Faruque, M. O., Rahaman, M. F., Hoque, M. A., Ikeya, K., Amano, T., Han, J., Dorji, T., & Omar, A. I. (2015). Present status of gayal (*Bos frontalis*) in the home tract of Bangladesh. *Bangladesh Journal of Animal Science*, 44(1), 75–84. <https://doi.org/10.3329/bjas.v44i1.23147>
- [6] Geng, Y., Hu, G., Ranjitkar, S., Wang, Y., Bu, D., Pei, S., Ou, X., Lu, Y., Ma, X., & Xu, J. (2017). Prioritizing fodder species based on traditional knowledge: A case study of Mithun (*Bos frontalis*) in Dulongjiang area, Yunnan Province, Southwest China. *Journal of Ethnobiology and Ethnomedicine*, 13, 24. <https://doi.org/10.1186/s13002-017-0153-z>
- [7] Geng, Y., Ranjitkar, S., Yan, Q., He, Z., Su, B., Gao, S., Niu, J., Bu, D., & Xu, J. (2020). Nutrient value of wild fodder species and the implications for improving the diet of Mithun (*Bos frontalis*) in Dulongjiang area, Yunnan Province, China. *Plant Diversity*, 42(6), 455–463. <https://doi.org/10.1016/j.pld.2020.09.007>
- [8] Giasuddin, M., & Islam, M. R. (2003). Physical feature, physiological character and behavior study of gayal (*Bos frontalis*). *Asian-Australasian Journal of Animal Sciences*, 16(11), 1599–1603. <https://doi.org/10.5713/ajas.2003.1599>
- [9] Gupta, N., Gupta, S. C., Verma, N. D., Pundir, R. K., Joshi, B. K., Nivsarkar, A. E., & Sahai, R. (1996). Mithun: An important bovine species of Indian origin. *Animal Genetic Resources Information*, 18, 41–48. <https://doi.org/10.1017/S1014233900000699>
- [10] Kumar, H., Choudhary, J., Chotso, K., Mech, M. M., Devi, L. S., Hanah, S. S., Chamuah, J. K., & Shivanagowda, G. P. (2025). Mithun (*Bos frontalis*): A forest-based food animal of the Eastern Himalayas. *Journal of Meat Science*, 20(2). <https://doi.org/10.48165/jms.2025.20.02.1>
- [11] Lambert, A. B. (1804). Description of *Bos frontalis*, a new species, from India. *Transactions of the Linnean Society of London*, 7(1), 57–59. <https://doi.org/10.1111/j.1096-3642.1804.tb00280.x>
- [12] Mondal M, Baruah K K and Rajkhowa C (2014): Mithun: An Animal of Indian Pride. *Livestock Research for Rural Development*. Volume 26, Article #6. Retrieved July 18, 2026, from <http://www.lrrd.org/lrrd26/1/mond26006.html>
- [13] Prakash, B., Dhali, A., Mech, A., Khate, K., Moaakum, H., & Rajkhowa, C. (2008). Effect of feeding forest foliages, rice straw and concentrate-based total mixed ration on nutrient utilization and growth in Mithun (*Bos frontalis*). *Livestock Science*, 117(2–3), 263–269. <https://doi.org/10.1016/j.livsci.2007.12.021>
- [14] Sentichuba Jamir and Neelam Khare (2018). Food, Feeding Habits of Mithun (*Bos frontalis*) in Protected Area of Nagaland. *Int.J.Curr.Microbiol.App.Sci*. 7(10): 1416-1426. <https://doi.org/10.20546/ijcmas.2018.710.157>
- [15] Shisode, M & Khanvilkar, A & Kulkarni, M & Samant, S & Yadav, G & Bawaskar, M. (2009). Mithun: The Pride animal of North-eastern hilly region of India. *Veterinary World*. 2(12.000): 480-481.
- [16] Sinha, D., Prakash, B., Neog, B. N., Baruah, K. K., Sarmah, S., & Rajkhowa, C. (2011). Intake, digestibility and nitrogen balance in Mithun (*Bos frontalis*) offered urea-treated paddy straw based feed blocks. *Tropical Animal Health and Production*, 43(2), 383–387. <https://doi.org/10.1007/s11250-010-9703-2>
- [17] Taba, R., Kimsing, A. T., Biju, T., Ngukir, J., Tamir, T., Tag, H., Payum, T., & Mize, D. (2015). Fodder plants of Mithun (*Bos frontalis* Lambert, 1804): A case study of West Siang, Arunachal Pradesh. *Journal of Bioresources*, 2(2), 74–78.